B. TECH-ME-IV-I Sem

23A03602T	**CAD/CAM (Professional Core)**	**L**	**T**	**P**	**C**
		3	**0**	**0**	**3**

Course objectives: The objectives of the course are to	
1	Understand the basic of CAD/CAM. Explore graphics standards and analyze 2D and 3D geometric transformations.
2	Knowledge and skills to apply various geometric modeling techniques, as well as solid modeling approaches.
3	Explain the principles of Computer Aided Manufacturing (CAM), numerical control (NC), and the functionalities of CNC and DNC systems.
4	Design and develop a part programming using G/M codes and APT for various machining operations.
5	Explain the basics of automation systems, robotics, group technology, CIM, and emerging trends like VR, AR, and AI.

Course Outcomes: On successful completion of the course, the student will be able to,		
1	Apply and Evaluate the CAD/CAM principles to design and manufacturing processes. Create and manipulate CAD models using appropriate software tools.	*L3,L4,L5*
2	Analyze and differentiate between Hermite, Bezier, and B-spline curves and construct and manipulate surface models. Evaluate the effectiveness of Boolean operations and create complex geometric models.	*L2,L3,L5, L6*
3	Identify and analyze key components of NC and CNC systems in manufacturing processes. Evaluate the performance of DNC systems in a manufacturing environment.	*L1,L2,L3, L4,L5*
4	Explain the structure and functioning of NC/CNC/DNC machine tools and adaptive control systems. Develop part programs using standard codes and APT for basic machining operations.	*L2,L6*
5	Understand and evaluate the role of robotics, group technology, and CIM in modern manufacturing systems, and explain emerging technologies like VR, AR, and AI.	*L2,L5*

UNIT –I

Overview of CAD/CAM: Product cycle, CAD, CAM and CIM. CAD Tools, CAM Tools, Utilization in an Industrial Environment, Evaluation criteria. CAD data structure, Data base management systems.

Computer Graphics: Co-ordinate systems, Graphics package functions, 2D and 3D transformations, clipping, hidden line / surface removal color, shading.

UNIT –II

Geometric Modeling: Representation techniques, Parametric and non-parametric representation, various construction methods, wire frame modeling, synthetic curves and their representations, surface modeling, synthetics surfaces and their representations. Solid modeling, solid representation, fundamentals, introduction to boundary representations, constructive solid geometry representations

UNIT- III

Numerical Control: NC, NC Modes, NC Elements, NC Machine tools and their structure, Machining center, types and features. Controls in NC, CNC systems, DNC systems. Adaptive control machining systems, types of adaptive control.

CNC Part Programming: Fundamentals, NC word, NC Nodes, canned cycles, cutter radius compensation, length compensation, computed assisted part programming using APT: Geometry statements, motion statements, post process statements, auxiliary statements, macro statement program for simple components.

UNIT -IV

Group Technology & FMS: Part Family, Classification and Coding, advantages & limitations, Group technology machine cells, benefits. FMS: Introduction, components of FMS, material handling systems, Computer control systems, advantages.
Computer Aided Quality Control: Terminology in Quality control, Inspection and testing, Contact inspection methods - optical and non-optical, integration of CAQC with CAD and CIM.

UNIT- V

Computer Aided Processes Planning: Retrieval type and Generative type, benefits Machinability data systems, Computer generated time standards.

Computer integrated production planning: Capacity planning, shop floor control, MRP-I, MRP-II, CIMS benefits. Trends in manufacturing systems: Concepts of Reconfigurable manufacturing, Sustainable manufacturing and lean manufacturing.

Textbooks:

1. Mikell P. Groover, Emory W. Zimmers , CAD/CAM, 5/e, Pearson Prentice Hall of India, Delhi, 2008.
2. Ibrahim Zeid, R.Siva Subramanian, CAD/CAM: Theory and Practice, 2/e, Tata McGraw-Hill, Delhi, 2009.

Reference Books:

1. P. N. Rao, CAD/CAM: Principles and applications, 3/e, Tata McGraw-Hill, Delhi, 2017.
2. P. Radhakrishnan, S. Subramanyan& V. Raju, CAD/CAM/CIM, 3/e, New Age International Publishers, 2008.
3. Computer Aided Manufacturing, 3/e, Tien Chien Chang, Pearson, 2008.

Online Learning Resources:

- https://onlinecourses.nptel.ac.in/noc20_me44/preview
- https://www.youtube.com/watch?v=EgKc9L7cbKc
- https://www.youtube.com/watch?v=KXFpTb9cBpY
- https://web.iitd.ac.in/~hegde/cad/lecture/L01_Introduction.pdf
- https://www.vssut.ac.in/lecture_notes/lecture1530947994.pdf
- https://www.iare.ac.in/sites/default/files/lecture_notes/CAD_CAM_LECTURE_NOTES.pdf